Linear algebra: Exercise chapter 10

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1. Find an orthogonal matrix Q that diagonalizes $S = \begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & -2 \\ 2 & -2 & 0 \end{bmatrix}$. What is the spectral decomposition of S?

   Answer. To find an orthogonal matrix Q that diagonalizes the matrix S and the spectral decomposition of S, we need to follow these steps:

   Step 1: Finding the eigenvalues of S:

$$\det(S - \lambda I) = 0 \Longrightarrow \begin{vmatrix} 1-\lambda & 0 & 2 \\ 0 & -1-\lambda & -2 \\ 2 & -2 & -\lambda \end{vmatrix} = (1-\lambda)\begin{vmatrix} -1-\lambda & -2 \\ -2 & -\lambda \end{vmatrix} - 0 + 2\begin{vmatrix} 0 & -1-\lambda \\ 2 & -2 \end{vmatrix} = \lambda(9-\lambda^2) = 0.$$

   Thus $\lambda_1 = 0, \quad \lambda_2 = 3, \quad \lambda_3 = -3.$

   Step 2: Finding the eigenvectors of S:

   $\lambda_1 = 0$:

$$(S - \lambda_1 I)v_1 = 0 \Longrightarrow Sv_1 = 0 \Longrightarrow \begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & -2 \\ 2 & -2 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Longrightarrow \begin{cases} x_1 + 2x_3 = 0 \\ x_2 + 2x_3 = 0 \\ 2x_1 - 2x_2 = 0 \end{cases} \Longrightarrow v_1 = \frac{1}{3}\begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}.$$

   $\lambda_2 = 3$:

$$(S - \lambda_2 I)v_2 = 0 \Longrightarrow (S - 3I)v_2 = 0 \Longrightarrow \begin{bmatrix} -2 & 0 & 2 \\ 0 & -4 & -2 \\ 2 & -2 & -3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Longrightarrow \begin{cases} -2x_1 + 2x_3 = 0 \\ 4x_2 + 2x_3 = 0 \\ 2x_1 - 2x_2 - 3x_3 = 0 \end{cases} \Longrightarrow v_2 = \frac{1}{3}\begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}.$$

   $\lambda_3 = -3$:

$$(S - \lambda_3 I)v_3 = 0 \Longrightarrow (S + 3I)v_3 = 0 \Longrightarrow \begin{bmatrix} 4 & 0 & 2 \\ 0 & 2 & -2 \\ 2 & -2 & 3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Longrightarrow \begin{cases} 4x_1 + 2x_3 = 0 \\ 2x_2 - 2x_3 = 0 \\ 2x_1 - 2x_2 + 3x_3 = 0 \end{cases} \Longrightarrow v_3 = \frac{1}{3}\begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}.$$

   Step 3: Form the orthogonal matrix Q and the diagonal matrix of eigenvalues Λ

$$Q = \frac{1}{3}\begin{bmatrix} 2 & 2 & -1 \\ 2 & -1 & 2 \\ -1 & 2 & 2 \end{bmatrix}, \qquad \Lambda = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -3 \end{bmatrix}.$$

$$S = Q\Lambda Q^T = \begin{bmatrix} \frac{2}{3} & \frac{2}{3} & \frac{-1}{3} \\ \frac{2}{3} & \frac{-1}{3} & \frac{2}{3} \\ \frac{-1}{3} & \frac{2}{3} & \frac{2}{3} \end{bmatrix}\begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -3 \end{bmatrix}\begin{bmatrix} \frac{2}{3} & \frac{2}{3} & \frac{-1}{3} \\ \frac{2}{3} & \frac{-1}{3} & \frac{2}{3} \\ \frac{-1}{3} & \frac{2}{3} & \frac{2}{3} \end{bmatrix}.$$

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2. Find all orthogonal matrices that diagonalize $S = \begin{bmatrix} 9 & 12 \\ 12 & 16 \end{bmatrix}$.

   Answer.

Solution

Finding eigenvalues and eigenvectors of S:

$$\det(S - \lambda I) = 0 \Longrightarrow \begin{vmatrix} 9-\lambda & 12 \\ 12 & 16-\lambda \end{vmatrix} = (9-\lambda)(16-\lambda) - (12)(12) = \lambda(\lambda - 25) = 0.$$

Thus, $\lambda_1 = 25, \quad \lambda_2 = 0.$

$\lambda_1 = 25$:

$$(S - \lambda_1 I)v_1 = 0 \implies (S - 25I)v_1 = 0 \implies \begin{bmatrix} -16 & 12 \\ 12 & -9 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies -4x_1 + 3x_2 = 0 \implies v_1 = \frac{1}{5}\begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

$\lambda_2 = 0$:

$$(S - \lambda_2 I)v_2 = 0 \implies Sv_2 = 0 \implies \begin{bmatrix} 9 & 12 \\ 12 & 16 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \implies 3x_1 + 4x_2 = 0 \implies v_2 = \frac{1}{5}\begin{bmatrix} 4 \\ -3 \end{bmatrix}.$$

All orthogonal matrices Q:

$$Q_1 = \begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ \frac{4}{5} & -\frac{3}{5} \end{bmatrix}, \qquad Q_2 = -\begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ \frac{4}{5} & -\frac{3}{5} \end{bmatrix} = \begin{bmatrix} -\frac{3}{5} & -\frac{4}{5} \\ -\frac{4}{5} & \frac{3}{5} \end{bmatrix}.$$

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3. Consider $A = \begin{bmatrix} 2 & b \\ 1 & 0 \end{bmatrix}$. What b makes $A = Q\Lambda Q^T$ possible? What values of b will make it impossible to diagonalize A? What values makes A singular?

Answer. For first question, since $A = Q\Lambda Q^T$ is symmetric, thus $b = 1$.

For second question, we have

$$\det(A - \lambda I) = \begin{bmatrix} 2-\lambda & b \\ 1 & -\lambda \end{bmatrix} = \lambda^2 - 2\lambda - b = 0 \implies \lambda = 1 \pm \sqrt{1+b}.$$

If $b < -1$, then it is impossible to diagonalize A.

For third question, if $b = 0$, then $\det(A) = 0$ and A is singular.

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4. Find $S \in \mathbb{S}^{2\times 2}$ such that 2 is the only eigenvalue of S. Is this matrix unique?

Answer. For $S \in \mathbb{S}^{2\times 2}$ (the set of symmetric 2×2 matrices), we can write:

$$S = \begin{bmatrix} a & c \\ c & b \end{bmatrix}.$$

We have

$$\det(S - \lambda I) = \begin{vmatrix} a-\lambda & c \\ c & b-\lambda \end{vmatrix} = (a-\lambda)(b-\lambda) - c^2 = \lambda^2 - (a+b)\lambda + (ab - c^2).$$

Since 2 is the only eigenvalue, the characteristic polynomial of S must be:

$$\det(S - \lambda I) = (\lambda - 2)^2 = 0 \implies \lambda^2 - (a+b)\lambda + (ab - c^2) = \lambda^2 - 4\lambda + 4.$$

Thus, $a + b = 4$ and $ab - c^2 = 4$ implying

$$a(4-a) - c^2 = 4 \implies c^2 + a^2 - 4a + 4 = 0 \implies c^2 + (a-2)^2 = 0 \implies c = 0, \quad a = b = 2.$$

Therefore, this matrix is unique and

$$S = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}.$$

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5. Which of the following matrices have two positive eigenvalues? Use tests, don't compute the eigenvalues. What does it mean for a matrix to be positive definite?

$$A = \begin{bmatrix} 5 & 6 \\ 6 & 7 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & -2 \\ -2 & -5 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 10 \\ 10 & 100 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 10 \\ 10 & 101 \end{bmatrix}.$$

Answer.

Matrix A: $A \in S^{2\times 2}$ and $\det(A) = -1$. Thus, A has two real eigenvalues: one positive and one negative. Hence, A is not positive definite.

Matrix B: $B \in S^{2\times 2}$, $\det(B) = 1$, and $\text{trac}(B) = -6$. Thus, B has two negative eigenvalues. Hence, B is not positive definite.

Matrix C: $C \in S^{2\times 2}$, $\det(C) = 0$, and $\text{trac}(C) = 101$. Thus, C has one positive eigenvalue and one zero eigenvalue. Hence, C is positive semidefinite.

Matrix D: $D \in S^{2\times 2}$, $\det(D) = 1$, and $\text{trac}(D) = 102$. Thus, D has two positive eigenvalues. Hence, D is positive definite.

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6. Choose a matrix M from previous exercise such that M is not positive (semi)definite. Find a vector x such that $x^T Mx < 0$.

Answer. The matrix B is not positive definite and

$$\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} -1 & -2 \\ -2 & -5 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = -1 < 0.$$

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7. Which one of the following matrix is positive (semidefinite) definite?

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 4 \\ 4 & 3 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -5 & 2 \\ -1 & 2 & 5 \end{bmatrix}, \quad D = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 3 & 2 \\ -1 & 2 & 4 \end{bmatrix}.$$

Answer. **Matrix A:** For each $x \in \mathbb{R}^2$ with $x \neq 0$, we obtain

$$x^T Ax = [x_1 \ x_2] \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1^2 + x_1x_2 + x_2^2 > 0.$$

Thus, A is positive definite.

Matrix B: $B \in S^{2\times 2}$ and $\det(B) = -7 < 0$, and $\operatorname{trac}(B) = 6$. Thus, B has two real eigenvalues: one positive and one negative. Hence, B is not positive (semidefinite) definite.

Matrix C: $C_{22} = -5$. Hence, C is not positive (semidefinite) definite.

Matrix D: $D \in S^{3\times 3}$.

$$\det(D - \lambda I) = 0 \Longrightarrow \begin{vmatrix} 2-\lambda & 0 & -1 \\ 0 & 3-\lambda & 2 \\ -1 & 2 & 4-\lambda \end{vmatrix} = (2-\lambda) \begin{vmatrix} 3-\lambda & 2 \\ 2 & 4-\lambda \end{vmatrix} - 0 - 1 \begin{vmatrix} 0 & 3-\lambda \\ -1 & 2 \end{vmatrix}$$
$$= -\lambda^3 + 9\lambda^2 - 21\lambda + 13 = 0.$$

Thus $\lambda_1 = 1, \quad \lambda_2 = 4 + \sqrt{3}, \quad \lambda_3 = 4 - \sqrt{3}$. All eigenvalues are positive, thus D is positive definite.

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8. True or false:

   (a) Every positive definite matrix is invertible.

   (b) A diagonal matrix with positive diagonal entries is positive definite.

   (c) A symmetric matrix with a positive determinant is positive definite.

Answer.

(a): True. If A is a positive definite matrix, then all eigenvalues of A are positive and consequently, determinant of A is positive. Thus, A is invertible.

(b): True. A diagonal matrix with positive diagonal entries is a symmetric matrix with positive eigenvalues (diagonal entries) making the matrix positive definite.

(c): False. A positive determinant means the product of the eigenvalues is positive, but it does not guarantee that all eigenvalues are positive. For example, the matrix

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

is symmetric and $\det(A) = 1 > 0$, but it is not positive definite because two eigenvalues are negative.

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9. Without multiplying $S = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix}$, find:

   (a) $\det(S)$.

   (b) The eigenvalues of S.

   (c) The eigenvectors of S.

   (d) A reason why S is positive definite.

Answer.

(a):

$$\det(S) = \begin{vmatrix} 2 & 0 \\ 0 & 5 \end{vmatrix} = 2 \times 5 = 10.$$

(b): $\lambda_1 = 2, \quad \lambda_2 = 5.$

(c):

$$v_1 = \begin{bmatrix} \cos(\theta) \\ \sin(\theta) \end{bmatrix}, \qquad v_2 = \begin{bmatrix} -\sin(\theta) \\ \cos(\theta) \end{bmatrix}.$$

(d): The matrix S is positive definite because it is symmetric, and all its eigenvalues are positive.